THE JOURNAL OF
PHYSICAL CHEMISTRY C

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J. Phys. Chem. C, **Just Accepted Manuscript** • DOI: 10.1021/acs.jpcc.7b07315 • Publication Date (Web): 13 Oct 2017

A Molecular Simulation Study of Carbon Dioxide Uptake by a Deep Eutectic Solvent Confined in Slit Nanopores

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Abstract

Molecular dynamics (MD) simulations were performed to study the behavior of CO_2 in varying amounts of a common deep eutectic solvent (DES), choline chloride and ethylene glycol (termed ethaline), confined in slit-like pores of width H = 5.2 nm with graphite or rutile walls at T = 318 K. In the absence of DES, CO_2 adsorbs to the pore walls, but increasing amounts of ethaline inside the pores quickly displace carbon dioxide into the gas/liquid interfaces, and into dissolution within the confined DES. This process is driven by strong interactions of the ethaline components with the pore walls, especially in the case of rutile systems, which also cause the local ratio of choline chloride : ethylene glycol inside the pores to depart from the bulk value of 1:2. As the amount of DES inside the pores increases, the diffusivity of CO_2 reaches a maximum in partially filled pores and decays to the values observed in pores filled with DES, which are similar to the diffusion coefficients of CO_2 in the bulk DES. The average number densities of CO_2 near confined ethaline (i.e., dissolved in the DES, and adsorbed at the pore walls and at the gas/liquid interface) are significantly larger than the corresponding value observed in bulk ethaline; for pores partially filled with DES, the overall average number density of CO_2 can be ~3.0-7.3 times the value observed in bulk ethaline. Even though larger number densities of CO_2 are observed in pores with no ethaline adsorbed, our results suggest that systems of nanoporous

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materials partially filled with DESs could be further explored and optimized for separation of carbon dioxide and other gases, in analogy to the development of supported ionic liquid phase materials for similar purposes.

1. Introduction

Deep eutectic solvents (DESs) usually consist of two or three components that associate with each other through hydrogen bonds. These interactions hinder the crystallization process, and therefore the mixture freezes at a temperature considerably lower than the freezing point of the pure components. First introduced by Abbott *et al.*,[1] DESs have recently attracted attention as promising alternatives to ionic liquids (ILs), as they share many of their physical and chemical properties while being environmentally friendlier and much cheaper and easier to prepare.[2] In analogy to ILs, different combinations of hydrogen bond donors (HBDs) and hydrogen bond acceptors (HBAs) can result in DESs with diverse properties. Therefore, DESs have been considered for a number of applications such as metal processing, chemical synthesis, solvents and separations,[3-6] including CO_2 capture.[7-10] However, in analogy to ILs many typical DESs are dense and viscous fluids, leading to mass transfer issues in CO_2 separations.[11,12] Transport limitations in systems of ILs for CO_2 capture can be overcome by using supported ionic liquid membranes and supported ionic liquid phase materials, in which micron-thick membranes or nanoporous materials containing small amounts of ILs are used for gas separations.[11-23] Very recently, Budhathoki *et al*. used molecular simulations to evaluate model supported ionic liquid membranes and supported ionic liquid phase material for CO_2 separations from CH_4 and H_2.[24] They found increased permselectivities (equal to the diffusion selectivity times the solubility selectivity) of CO_2 when using slit carbon pores completely filled with the IL 1-*n*-butyl-3-

methylimidazolium bis(trifluoromethylsulfonyl)imide, [C_4mim^+][Tf_2N^-], as compared to empty slit pores, but the diffusivity of CO_2 in the pores filled with the IL was significantly smaller than the value observed in the empty carbon pore.

Motivated by the idea of supported IL porous materials for CO_2 capture, here we present a molecular simulation study of the behavior of CO_2 in a typical DES confined inside slit-like nanopores with graphite or rutile walls. Here we also explore the effect of different amounts of DES inside the pores on the properties of carbon dioxide. Another motivation for our study is the 'oversolubility' behavior reported in recent experimental and simulation studies,[25-30] where increased gas solubility was found in liquids confined in nanoporous materials as compared to the values observed in the bulk liquids. Ho *et al*.[27] have described three molecular mechanisms for gas uptake in a solvent confined in nanopores: (1) surface adsorption, driven by the interaction between the gas molecules and the pore surfaces; (2) confinement-induced enhanced solubility, in which gas uptake is favored in the regions of low solvent density formed by the layering of the solvent, and (3) adsorption at the gas/liquid interface in pores partially filled with solvent. A fundamental question asked here is how confinement in nm-sized pores affects the behavior of carbon dioxide in DESs. Our group have previously used molecular dynamics simulations to study the behavior of a confined DES (choline iodide + glycerol), finding significant differences in the structural and dynamical properties when the DESs is confined in slit pores with graphite or rutile walls.[31] Here we have chosen to study the behavior of carbon dioxide in a different DES, composed of choline chloride and ethylene glycol with a 1:2 molar ratio (termed as ethaline thereafter, see Figure 1). Ethaline has a melting point of 207 K,[32] its viscosity (37-41 mPa·s at 298 K[33,34]) is about one order of magnitude lower when compared to other choline chloride-based DESs,[33,34] and the experimental solubility of carbon dioxide in ethaline is around 3.13 mol/kg at 303 K and 5.9 MPa.[35] The rest of our paper is organized as follows: Section 2 contains details of the computational models and methods used in this study, followed by our main results presented and discussed in Section 3, and in Section 4 the main conclusions of our work.

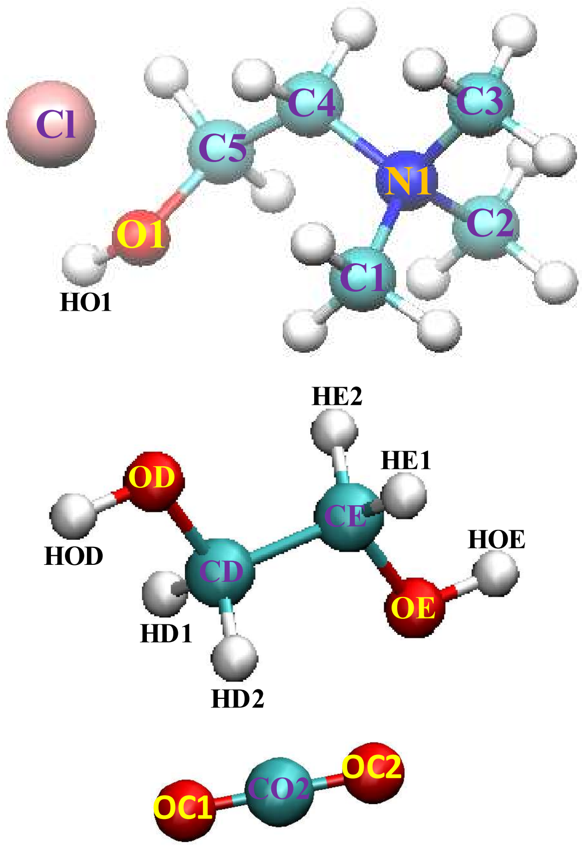

Figure 1. Schematic representation of the molecules of the ethaline components choline chloride (top) and ethylene glycol (center), and carbon dioxide (bottom). The labels are atom notations used in this work.

2. Computational details

We performed classical molecular dynamics simulations of the ethaline DES, which consists of choline chloride and ethylene glycol at a molar ratio of 1:2, with and without carbon dioxide. We modeled these systems confined inside slit nanopores, where the walls are made of graphite or TiO_2 [rutile(110)]; we also performed simulations of these systems in the bulk and using elongated boxes (see below). Details of our model systems are presented in Table 1, and representative snapshots for most of our model systems at equilibrium are shown in Figures 2 and S1 (Supporting Information). The simulation setup for our confined systems is the same used in our previous study,[31] in which both sides of the slit pore in the *x* direction are connected to two reservoirs (Figs. 2 and S1) where all molecules are placed in our initial configurations; as described below, we ran separate simulations of the CO_2 in the bulk reservoirs to provide estimates of the bulk pressure outside of our pores. We considered varying amounts of ethaline inside the pores of our confined systems (Table 1, Figs. 2 and S1). The total number of choline cations, chloride anions, and ethylene glycol and CO_2 molecules was arbitrary, but very similar to the amounts and ratios used by García *et al*.[7] We also ran a few systems with larger total amounts of carbon dioxide molecules, as to obtain bulk gas pressures that were between 11-18 bar for all systems (see below). Our MD simulation setup and approach for confined systems can provide reasonable estimates of equilibrium properties of all species in our systems (see discussion at the end of this section). Rigorously speaking, under equilibrium conditions the temperature and chemical potential of CO_2 inside a pore containing a given amount of ethaline and in the bulk phase must be equal. This criterion yields the equilibrium amount of CO_2 inside the pore, at any given values of temperature and chemical potential (which can then be related to the pressure) in the bulk gas. Therefore, simulations in the grand canonical ensemble would allow one to rigorously determine equilibrium properties for these systems. We intend to consider this type of simulations, in combination with other calculations in our future studies in this area (see Conclusions).

In Table 1, GC1, GC3, GC4, GC8 indicate graphite pores with ethaline and CO_2 present, while TiC1, TiC3, TiC4 and TiC8 indicate rutile pores containing both ethaline and CO_2. The number is a short-hand notation equal to the number of ion pairs (i.e., choline chloride) divided by 100, in every particular system. Similarly, systems G4, G8, Ti4 and Ti8 are graphite and rutile systems without CO_2. Systems LC3, LC4, L3 and L4 represent elongated boxes (no pore walls) containing ethaline with CO_2 (LC3, LC4) or without CO_2 (L3, L4). Systems GC0a, TiC1a and TiC0a are similar to systems GC0, TiC1 and TiC0 but had a larger amount of CO_2 molecules; and system TiC4s is similar to TiC4, but has a pore that is shorter in the *x* direction. All our simulations were performed at a temperature T = 318 K using GROMACS 5.0.[36,37] The initial position of the atoms in choline chloride and ethylene glycol were obtained using PRODRG Server.[38] The force field parameters for choline chloride and ethylene glycol were obtained from the work of Perkins *et al.*,[39] where the bonded and nonbonded terms were taken from the general AMBER force field (GAFF),[40,41] and partial charges were set to a value of ±0.9e for the ions.[39,42] This set of partial charges and force field parameters can reproduce several properties of ethaline in the bulk in good agreement with experimental data.[39] Carbon dioxide was modeled using the TraPPE force field.[43] Each of the graphite nanopore walls consists of three layers of graphene sheets that were generated by Visual Molecular Dynamics (VMD),[44] which was also used as the visualization tool in our study. The rutile surface is a three-atom layer of TiO_2 built by repeating its unit cell in [100], [010], and [001] directions and cutting along the (110) plane. The force field parameters used for atoms in rutile and graphite walls are the same as in our previous work.[31] The positions of the atoms in the walls were kept fixed throughout the simulations.

All our slit pores in our confined systems had a width of H = 5.2 nm, which is the same size considered in our previous studies involving DESs[31] and ILs.[45,46] The surface area of our slit pores is 10.0 nm and 5.1 nm in the x and y directions respectively. The slit pores are placed in the center of a simulation box of dimensions 30.0 nm, 5.1 nm, and 13.5 nm in the x, y and z directions, with auxiliary walls (modeled as graphene layers) placed to prevent the diffusion of molecules into the vacuum regions that are on top and bottom of the slit pores in the z direction (Figs. 2 and S1). In addition, we modeled a rutile slit pore system with a smaller surface area, i.e., 6.0 nm in the x direction (TiC4s in Table 1; see also Fig. S1g). The motivation for considering such a system is as follows. As described in our previous study,[31] our rutile pores exhibit end-wall effects in the y-z plane, because one of the ends of the rutile walls has a larger local concentration of O atoms as compared to the other end, which in turn has a larger amount of Ti atoms (Ti and O has partial charges, and our rutile walls are electrically neutral). Therefore, one end of the rutile walls preferentially attracts the cations while the other end interacts strongly with the anions (Figs. 2 and S1). These end-wall effects, combined with strong interactions between the ethaline components and the rutile surfaces, cause the DES in the TiC4/Ti4 systems (where the amounts of ethaline and CO_2 are not enough to completely fill the pore) to accumulate at both sides of the pore, creating an nm-sized 'bubble' in the center of the pore (Figs. 2g, S1c). Similar bubbles were also observed for slit rutile pores partially filled with the IL [$Emim^+$][$TFMSI^-$].[46] We performed additional simulations considering the same amount of ethaline as in TiC4 and Ti4, but having a shorter rutile pore length in the x direction (TiC4s system in Table 1). Here ethaline reached the center of the pore, forming nm-sized concave 'menisci' at both sides of the pore (Fig. S1g). In our previous study,[31] to eliminate end-wall effects in rutile systems we only considered ions/molecules within the central 6 nm range in the *x*-direction of the pore for our analysis. However, here the small amounts of ethaline and CO_2 within the pores in some of our systems did not allow us to exclude ions/molecules at the end of the rutile pores (see, e.g., Figs. 2e-f). Therefore, here we considered all ions/molecules within our slit rutile and graphite pores when analyzing our simulation trajectories; we also considered ethaline and CO_2 in the gas/liquid interfaces, even when these are outside of the pore area (see, e.g., Figs. 2d-h). We also note again that in all of our simulations of confined systems (except system TiC4s), we considered pore walls that are intentionally longer in the *x* direction (10.0 nm) as compared to in the *y* direction (5.1 nm), in order to minimize end-wall effects by the rutile walls on the properties of ethaline and CO_2 adsorbed in the center of the pore.

Table 1. Details of our model systems. Subscripts in bulk gas pressure represent uncertainties in the last digit

Model system	**Wall material**	**Number of ion pairs + ethylene glycol molecules**	**Number of carbon dioxide molecules**	**Observed density of CO_2 in the reservoirs (molecule/nm^3)**	**Bulk gas pressure in confined systems (bar)**
GC1	Graphite	100 + 200	400	0.254	11.4_7
GC3	Graphite	300 + 600	400	0.274	13_1
GC4	Graphite	400 + 800	400	0.300	14_1
GC8	Graphite	800 + 1600	400	0.368	18_1
GC0	Graphite	0 + 0	400	0.105	4.1_3
GC0a	Graphite	0 + 0	1000	0.330	15_1
G4	Graphite	400 + 800	0	-	-

G8	Graphite	800 + 1600	0	-	-
TiC1	Rutile	100 + 200	400	0.107	4.9_3
TiC1a	Rutile	100 + 200	700	0.321	16_1
TiC3	Rutile	300 + 600	400	0.270	13_1
TiC4	Rutile	400 + 800	400	0.280	13_1
TiC4s (*)	Rutile	400 + 800	400	0.284	13_1
TiC8	Rutile	800 + 1600	400	0.344	16_1
TiC0	Rutile	0 + 0	400	-(**)	- (**)
TiC0a	Rutile	0 + 0	3000	0.275	13_1
Ti4	Rutile	400 + 800	0	-	-
Ti8	Rutile	800 + 1600	0	-	-
LC3	-	300 + 600	300	-	-
LC4	-	450 + 900	300	-	-
L3	-	300 + 600	0	-	-
L4	-	450 + 900	0	-	-
Bulk	-	300 + 600	0	-	-
Bulk(C)	-	300 + 600	21	-	-

(*) This system has a shorter slit pore with a length of 6.0 nm in the *x* direction.
(**) All CO_2 molecules in this system adsorbed in the rutile pore (no CO_2 molecules in the reservoirs at equilibrium)

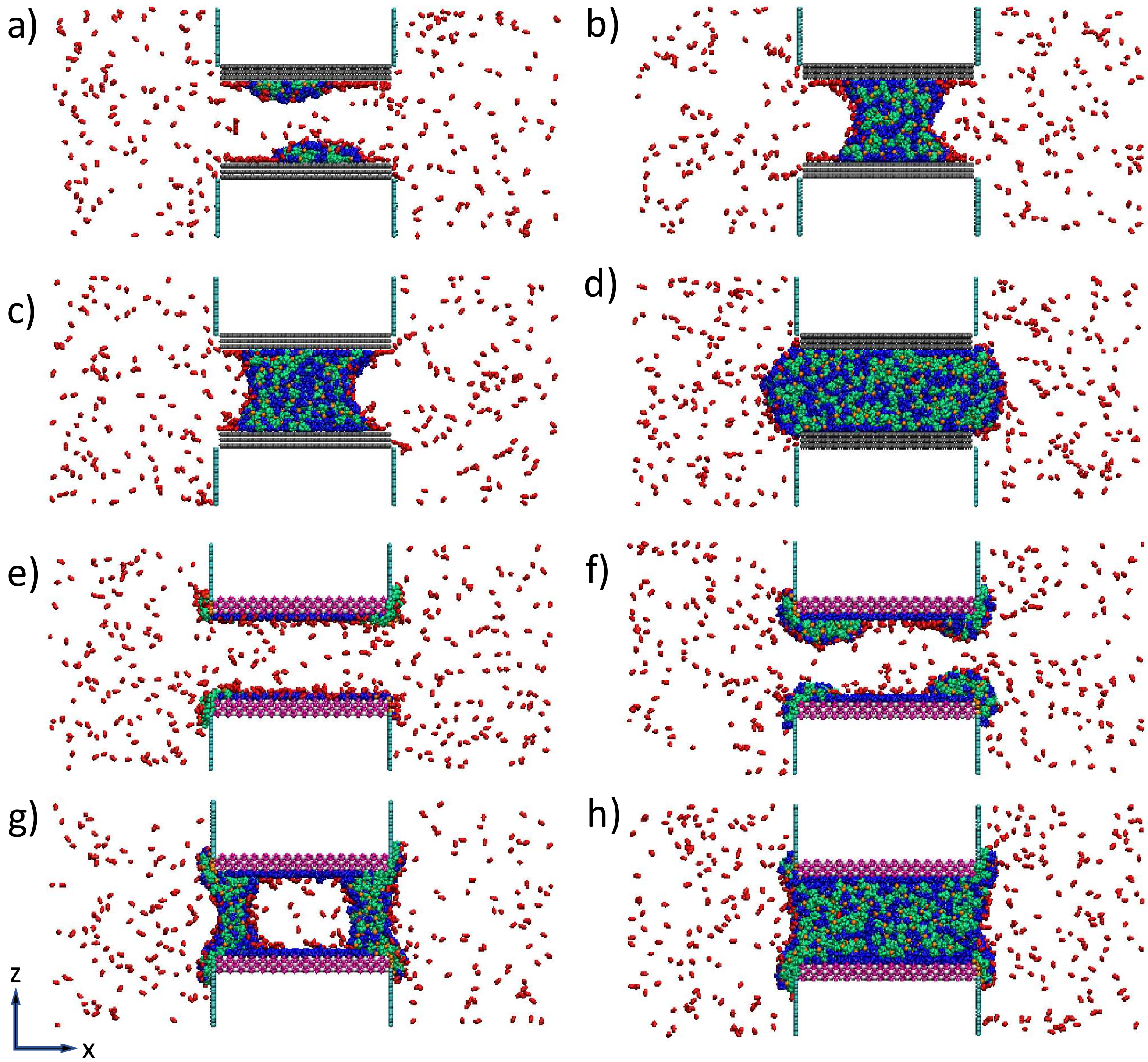

Figure 2. Representative simulation snapshots of model systems (a) GC1 (b) GC3 (c) GC4 (d) GC8 (e) TiC1a (f) TiC3 (g) TiC4 (h) TiC8 at equilibrium at T = 318 K. Molecules/atoms are colored as follows: DES: green = cation (choline), orange = anion (chloride), blue = HBD (ethylene glycol); red = carbon dioxide (CO_2). Rutile walls: white = Ti, magenta = O. Graphitic walls: grey = carbon. Auxiliary walls are colored in cyan.

In order to provide estimates of the pressure of the bulk gas phase outside of the pore in all our systems, we decided to conduct additional MD simulations in the *NVT* ensemble for pure CO_2 in the bulk at $T = 318$ K, at densities that are equal to the average density of CO_2 in the reservoirs at both sides of the pore in each of our systems (the observed gas densities in the reservoirs are reported in Table 1). We note that these reservoirs are relatively large, with a total volume of 20 nm × 5.1 nm × 13.5 nm, keeping in mind that the reservoirs are connected by periodic boundary conditions in all three directions (Figure 2), and only CO_2 is present in the reservoirs after equilibration of our pore systems. In estimating the bulk gas density in rutile systems, we only considered CO_2 molecules that are relatively far from the DES (i.e., we did not consider the CO_2 molecules adsorbed at the gas/DES interfaces that might be outside of the pore, e.g., see Figure 2d-h). These additional simulations yield a reasonable estimate of the external pressure of bulk CO_2 outside of the pore in our systems; these pressure values, which were obtained using standard tools from GROMACS, are reported in Table 1. The pressures determined via this approach range between 4.1 and 18 bar, and are similar to the gas pressures considered by Budhathoki *et al.*[24] for the case of CO_2 adsorption in confined ILs. As systems GC0, TiC0 and TiC1 exhibited bulk gas pressures that were low compared to those observed in the rest of our systems (Table 1), we performed additional MD simulations of these systems having a larger number of CO_2 molecules (systems GC0a, TiC0a and TiC1a, Table 1 and Figs. 2, S1); these systems have bulk gas pressures ranging between 13 and 16 bar, and thus are comparable to those observed in the rest of our systems (Table 1).

In order to obtain an approximate estimate of the amount of CO_2 that would be dissolved in bulk ethaline, we also performed simulations where a liquid phase of ethaline with CO_2 is in contact with vacuum regions (mimicking gas phases) in an elongated simulation box of dimensions 4.5 nm × 4.5 nm × 40 nm in the *x*, *y* and *z* directions (systems LC3/LC4 in Table 1; see also Fig. S1h). A rigorous determination of solubility isotherms of CO_2 in ethaline, which is a dense system involving ions and extensive hydrogen bonding, would require the use of techniques such as isothermal-isobaric Gibbs ensemble Monte Carlo coupled with configurational bias methods,[24,47] or continuous fractional component Monte Carlo.[48-52] However, here our intention was to compare the amounts of CO_2 present in confined and in bulk ethaline, and thus we decided to perform these simple MD simulations in an elongated box. We note that in future studies we could couple MD simulations with rigorous MC simulations of bulk and confined systems using the techniques mentioned above, combined with a computational strategy to optimize our choices of nanoporous materials and DESs/ILs/solvents as to maximize CO_2 separation (here we just used a common DES in common slit pores). Simulations with the LC3/LC4 systems suggest that a liquid ethaline system with 300 ion pairs and 600 molecules of ethylene glycol dissolves 21 molecules of CO_2 (not counting those molecules at the gas/liquid interface, as bulk systems typically have a low surface to volume ratio). Therefore, we also ran simulations of this liquid mixture in a cubic simulation box (i.e., without vacuum regions) to obtain properties for this liquid system [bulk(C) system in Table 1]. We also considered a bulk system of ethaline without CO_2 (bulk system in Table 1) to estimate properties for comparison purposes. Finally, we also modeled systems of just ethaline in contact with vacuum regions in an elongated box (L3/L4 in Table 1), in order to obtain insights into the structure of the gas/liquid interface of ethaline, and compare with previous results obtained for other DESs.[7]

Packmol[53] was used to place ethaline and CO_2 randomly for our initial configurations. The initial configurations were then subjected to an energy minimization procedure using the steepest descent method. After minimization, the systems were initially run at T = 500 K for 5 ns, and then the systems were annealed to 318 K over 2 ns and equilibrated at this temperature for 5 ns, followed by a 100 ns-long production. Finally, the above steps (starting from running 5 ns at T = 500 K) were repeated starting from the configuration obtained at the end of the first production run, resulting in two independent simulation runs for each system described in Table 1; all our results were averaged over these two independent realizations of each system. The improved velocity-rescaling thermostat of Parrinello *et al*.[54] was used to control the temperature in our simulations. We used a time step of 1 fs in all our simulations, and all bonds with hydrogen atoms were converted to constraints using LINCS algorithm.[55] The nonbonded interaction cutoff was equal to 1.5 nm for both Lennard-Jones and coulombic interactions. The particle-mesh Ewald (PME) method[56] was applied to calculate long range electrostatic interactions with a grid spacing of 0.1 nm and interpolation order of 6.

We note that here we have used the same simulation setup and methodology as in our previous MD study of DESs inside nanopores.[31] In that study, we presented rotational correlation functions that decay to zero for the cations and the HBD species inside the nanopores; the only correlation function that did not decay to zero within the timescale of our simulation was for the HBD near the rutile walls, which exhibited very slow dynamics due to extensive hydrogen bonding with the pore walls. These results suggest that our modeling approach leads to proper equilibration of our systems, even though metastabilities are ubiquitous in systems confined in nanopores, especially if the confined fluid is a DES (which are dense, have sluggish dynamics and exhibit strong electrostatic interactions and extensive hydrogen bonding). Although the DES examined in ref. 31 (choline iodide + glycerol) is not the same as the DES considered here (choline chloride + ethylene glycol), we do note that (1) ethaline is less viscous than choline iodide + glycerol; (2) as shown in Section 3.4, the presence of CO_2 lead to faster dynamics in these systems (CO_2 was not present in the work described in ref. 31); and (3) our simulations with ethaline were conducted for longer times (100 ns) compared to our simulations in ref. 31 (20 ns). Therefore, we are confident that our MD simulation approach can be used to provide reasonable estimates of equilibrium properties of all species in our systems.

3. Results and discussion

3.1. Local and average densities

The local number density profiles of each of the components of ethaline and CO_2 along the *z*-direction are presented in Figure 3 for most of the graphitic and rutile pore systems containing CO_2. For completeness, in Figure S2 (Supporting Information) we provide the number density profiles for those pore systems not included in Fig. 3, and in Figure S3 we present number density profiles for our simulations with elongated boxes. From simulations performed for bulk ethaline, we determined the average number densities of each component at 318 K to be 2.52 molecule/nm^3 for the choline cations and the chloride anions, and 5.04 molecule/nm^3 for ethylene glycol. For ease of visualization, the number density of HBD shown in Figs. 3, S2 and S3 is divided by 2 (since the ratio between ethylene glycol and choline chloride is 2). We first discuss results obtained for bulk systems in elongated boxes. When ethaline is in contact with vacuum (systems L3 and L4, see Fig. S3c-d), ethylene glycol tends to lie closer to the gas phase as compared to choline and chloride, which is similar to observations reported in simulation studies for other choline chloride-based DESs.[7] The accumulation of HBD at the interfaces leads to slightly smaller number density values for ethylene glycol within the liquid phase. When ethaline is in contact with carbon dioxide (systems LC3 and LC4, Figs. S3a-b and S1h), CO_2 tends to accumulate closer to the gas side of the interface, followed by HBD and the ions. The density profiles of the ethaline components in these systems are qualitatively similar to those observed for systems L3 and L4, which have no CO_2. Both LC3 and LC4 systems show a similar number density of CO_2 within the liquid phase, 0.176 molecule/nm^3 (Figs. S3a-b).

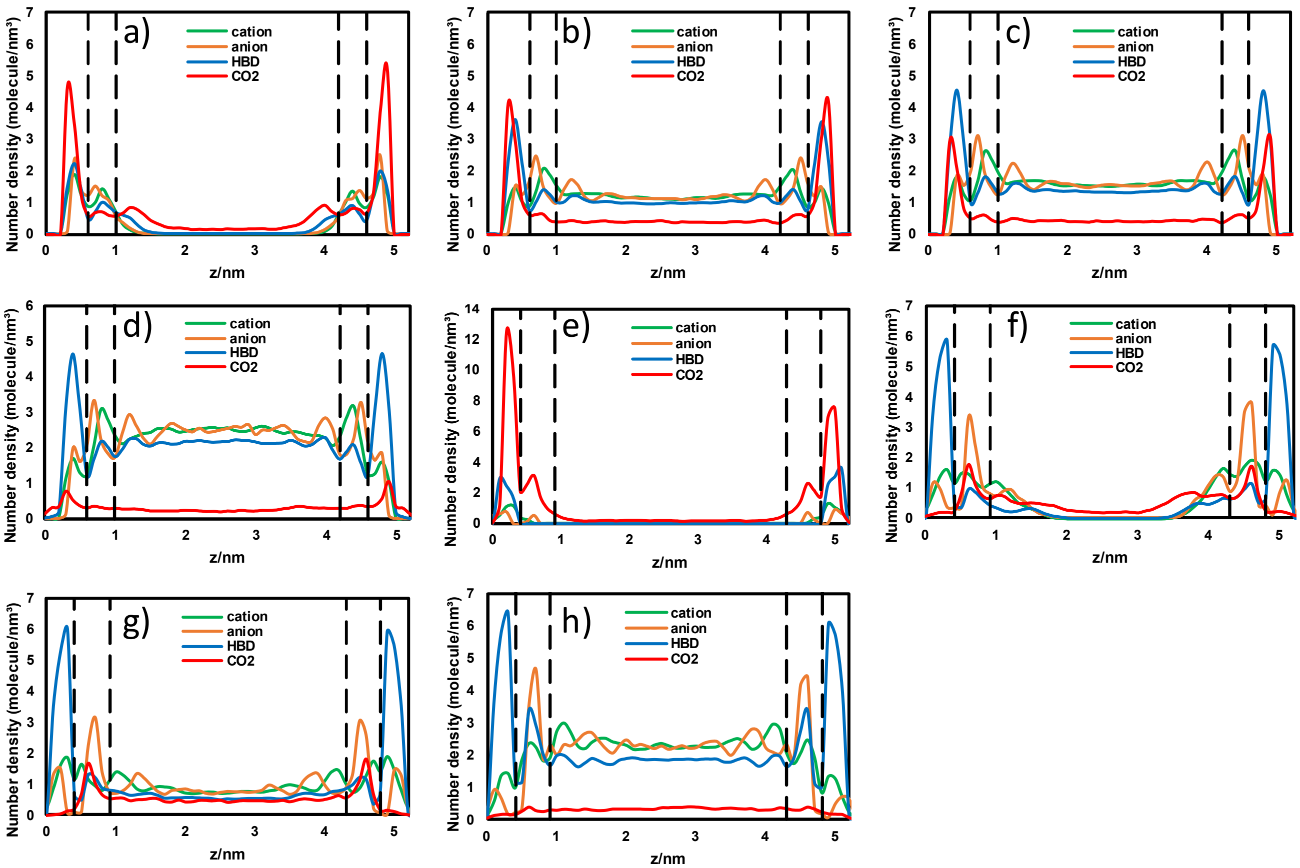

Figure 3. Number density profiles of ethaline species and CO_2 along the *z*-direction for (a) GC1, (b) GC3, (c) GC4 (d) GC8 (e) TiC1a (f) TiC3 (g) TiC4 (h) TiC8. Green = cation, orange = anion, blue = ethylene glycol (HBD), red = CO_2. The dashed lines represent borderlines between different regions inside our pore systems (first layers, second layers and center region). For ease

of visualization, the number density of ethylene glycol shown is the actual density divided by 2 (since the ratio between HBD and the ions is 2).

We now discuss the number density profiles observed for confined systems. As done in our previous study,[31] for our analysis we divided all species into three regions according to their *z*-distance from the pore walls (as suggested by the layering behavior of ethaline species shown in Figs. 3 and S2, see dashed lines): (1) first layers (first density peaks close to the pore walls), (2) second layers (next to the first layers) and (3) center region. For graphite pore systems that are fully filled with ethaline (system GC8, see Figs. 2d, 3d), ethylene glycol is the major component present in the 0.6 nm-thick first layers, with smaller amounts of cations, anions and CO_2 present. The number density of choline and chloride is larger in the 0.4 nm-thick second layers and in the 3.2 nm-thick center region, as compared to the densities observed for these species in the first layers; in contrast, the HBD exhibits lower number densities in the second layers and center region when compared to its density in the first layers. The local density of CO_2 exhibits a small peak in the first layer close to the graphitic walls, and very low number densities (~0.094 molecule/nm^3) are observed in the second layers and center region; this number is smaller than the equivalent quantity observed for CO_2 within bulk ethaline (~0.176 molecule/nm^3 in systems LC3 and LC4). In addition to being dissolved within confined ethaline in system GC8, CO_2 also adsorbs on the convex gas/liquid interfaces outside of the pore volume (Fig. 2d). As the amount of ethaline inside the graphitic pores decreases, the DES forms concave, nm-sized 'menisci' with the gas phase within the pore (systems GC4 and GC3, Figs. 2b and 2c), and in system GC1 only small amounts of liquid ethaline are adsorbed on the pore walls with a gas-like phase in the center of the pore (Fig. 2a). As the amount of ethaline inside the pore decreases, CO_2 exhibits increasingly larger densities in the first layers (Figs. 3a-c); furthermore, systems GC1, GC3 and GC4 exhibit larger number densities of CO_2 in the first and second layers and center regions as compared to system GC8. These observations are caused by a combination of the following two effects (Figs. 2a-c): (1) reductions in the amount of ethaline inside the pore leaves larger areas of the graphite surfaces available for adsorption of CO_2; and (2) carbon dioxide also tends to adsorb on the interfaces formed between liquid ethaline and the gas regions within the pore. In system GC1 the density peaks of choline and chloride in the first layers become taller than those observed in the second layers (compare Figs. 3a and 3b), but as the amount of ethaline inside the graphite pores is raised, increasing quantities of ethylene glycol are adsorbed on the graphite walls, reducing the densities of ions and CO_2 in the first layers (Figs. 3a-d).

For the rutile systems that are completely filled with ethaline (systems TiC8 and Ti8), ethylene glycol is the most dominant component in the 0.4 nm-wide first layers, with only a few ions present (Figs. 3h and S2d). Each molecule of ethylene glycol has two hydroxyl groups, which can form hydrogen bonds with the oxygen atoms at the rutile surfaces; these strong interactions between the HBD and the rutile walls drive most of the other species out of the first layers in these systems. Choline has one hydroxyl group and thus can also form hydrogen bonds with the oxygen atoms at the walls, especially near the edges of the rutile walls that are rich in O atoms; similarly, most of the anions in the first layers adsorb at the Ti-rich end side of the TiO_2 walls. This behavior is very similar to what we observed for choline iodide and glycerol inside a slit rutile pore.[31] Reductions in the total amount of ethaline in our systems lead to slight increases in the densities of both ions in the first layers (Figs. 3e-g). In contrast to the behavior observed as we reduce the amount of ethaline in graphite pores (Figs. 2a-d), the Ti-rich and O-rich end sides of the rutile walls make ethaline to accumulate at both ends of the rutile pores in systems TiC4/Ti4, making the DES to form liquid ‘bridges’ at the entrances of the pore (Figs. 2g and S1c). Ethylene glycol tends to spread all over the pore surfaces in the partially filled rutile systems TiC1a, TiC3 and TiC4 (Figs. 2e-g and 3e-g), and therefore exhibits the largest density among ethaline species in the first layers. As the amount of ethaline is reduced in the rutile systems, small density peaks appear for CO_2 in the second layers, with negligible amounts in the first layers (Figs. 3f-h), and in system TiC1a most of the CO_2 is adsorbed near the rutile walls (Fig. 3e). If we compare systems TiC1a (with a total of 700 CO_2 molecules, bulk gas pressure = 16 bar; Fig. 3e) and TiC1 (400 CO_2 molecules, bulk gas pressure = 4.9 bar, Fig. S2e), we note that the increase in CO_2 pressure lead to larger amounts of this species in the pore, with taller density peaks in the first and second layers for system TiC1a (compare Figs. 3e and S2e).

We note that for system TiC4, if we reduce the length of the rutile surfaces in the *x*-direction (system TiC4s, Fig. S1g), a behavior similar to that observed for system GC4 is observed, in which ethaline forms concave menisci at the sides of the pore. If we had considered a system equivalent to TiC4 but with a larger TiO_2 surface in the *x* direction while keeping the total amounts of ethaline and CO_2 constant, we would have obtained a system similar to the one illustrated in Figure 2g for system TiC4, only with a larger gas-like region in the center. The results obtained for systems TiC4 and TiC4s suggest that the pore blocking effects observed in the former system are not solely caused by the end-wall issues present in our rutile walls (see Section 2), but by a combination of these end-wall effects with strong interactions between ethaline and the rutile walls. We note that similar gas-like regions surrounded by liquid-like ‘bridges’ of adsorbate were observed in our previous simulation study of slit rutile pores partially filled with the IL [Emim$^+$][TFMSI$^-$] (see Figure 3a in ref. 46). In that study, we used a setup where the slit pore is not directly connected to reservoirs and thus has no auxiliary walls, and has periodic boundary conditions in the *x* and *y* directions; therefore, the rutile walls did not exhibit end-wall issues leading to possible 'hot spots' for adsorption, but liquid-like 'bridges' of adsorbate surrounded by gas-like regions were observed nevertheless in those systems. Pore blocking effects are known to influence adsorption and desorption experiments,[57] where a sample of the nanoporous material is typically immersed in the bulk adsorbate. For example, one could have a material where larger pores in the center of the adsorbent are connected to the bulk adsorbate phase through narrow pores. In this situation, the adsorbate could fill the narrow pores and block the entrance to the center pores, if the interactions between the adsorbate and the pore walls are very strong, and if the bulk pressure is not large enough to force the adsorbate to fill all pores. Pore blocking effects also affect desorption; one could have a situation where the nanoporous material is completely filled with adsorbate, and now the material in the narrow pores hinders the desorption of adsorbate from the pores in the center of the adsorbent.

The average number densities of CO_2 measured in the whole pore and in the different regions of the pore systems are presented in Figure 4. Two average number densities of CO_2 are reported, (1) within liquid ethaline inside the pore, but not accounting for CO_2 at the gas/liquid interfaces (Fig. 4a; here we focused on CO_2 dissolved within ethaline in our pore systems, and thus we did not consider systems GC1, TiC1, TiC1a and TiC3 due to the very small amounts of ethaline within the pores), and (2) accounting for all CO_2 near confined ethaline (Fig. 4b, i.e., carbon dioxide directly adsorbed at the pore walls, dissolved within ethaline, and adsorbed at the gas/liquid interfaces, even if the menisci are outside of the pore, e.g., Figs. 2d-h). Tables S1 and

S2 (Supporting Information) contain the data shown in Figures 4a and 4b respectively, as well as additional results not depicted in Fig. 4b. The number density of CO_2 within bulk ethaline, 0.176 molecules/nm^3, was estimated from the elongated simulation systems (LC3/LC4), and is also shown for reference in Fig. 4. Looking first at Fig. 4a, in general the overall average number densities of CO_2 dissolved in ethaline in our pore systems are smaller than the value observed in bulk ethaline. In general, for all systems the overall average number densities of CO_2 decrease with increasing amounts of ethaline inside the pore, being the lowest when the graphite or rutile pores are completely filled with ethaline (systems GC8 and TiC8). For all graphite systems, the average number density of CO_2 is largest in the first layers, almost doubling the values observed in the other regions of the pore; the center region has a slightly higher CO_2 density than that in the second layers. For the rutile systems TiC4, TiC4s and TiC8, the overall number density of CO_2 is smaller than what was observed for graphite systems with the same loading of ethaline. Negligible amounts of CO_2 are present in the first layers in rutile systems, and the results for systems TiC8 and TiC4s suggest that CO_2 has a larger density in the second layers than in the center regions (system TiC4 show an opposite trend, which could be caused by the gas-like region formed at the center of the pore, see Fig. 2g, which entraps significant amounts of CO_2).

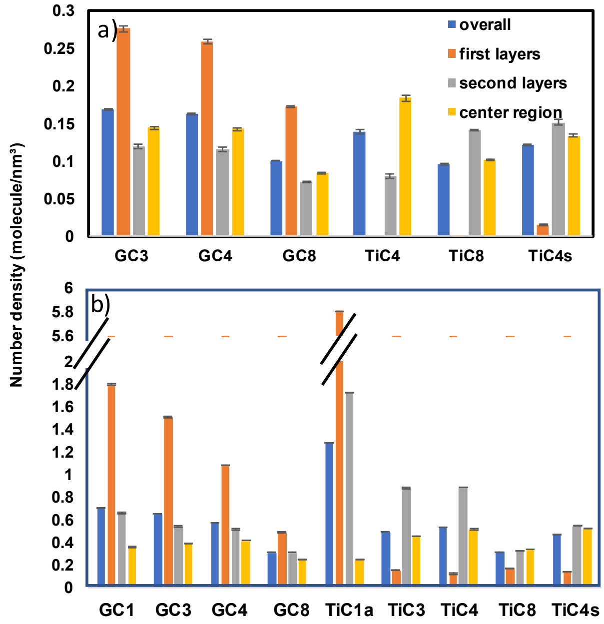

Figure 4. Average number density of carbon dioxide in the whole pore and in different regions of selected pore systems, for (a) CO_2 dissolved within ethaline inside the pore, and (b) all CO_2 near ethaline (i.e., dissolved in ethaline and adsorbed at the pore walls and at the gas/liquid interfaces). The dashed line indicates the average number density of carbon dioxide in the bulk of ethaline in the elongated systems (0.176 molecules/nm^3). Tables S1 and S2 (Supporting Information) contains the data shown in Figures 4a and 4b respectively, as well as additional results not depicted in Fig. 4b.

The snapshots shown in Fig. 2 suggest that important quantities of CO_2 adsorb at the gas/ethaline interfaces, and this fact is reflected in the results shown in Fig. 4b, where we report the average number density of all CO_2 near confined ethaline. In general, results shown in this figure indicate that if we account for CO_2 adsorbed at the pore walls, dissolved within ethaline, and adsorbed at the gas/liquid interfaces (even if the menisci are outside of the pore), the overall average number density of CO_2 is larger than the value observed for carbon dioxide dissolved in bulk ethaline.

For pores completely filled with ethaline (systems GC8 and TiC8), the difference between the results shown in Figs. 4a-b represents the CO_2 adsorbed at the gas/liquid interfaces, which is enough to push the average number density of carbon dioxide above the value observed for bulk ethaline. Even larger differences are observed in the partially filled pore systems; the overall average number density of CO_2 in graphite systems can be ~3-4 times the value observed in bulk ethaline, and up to 7.3 times the bulk value for the case of rutile systems. For graphite systems, the average number density of CO_2 in the first layers is significantly larger than the values observed for the second layers, which in turn are larger than those in the center region. In graphite systems partially filled with ethaline, large amounts of CO_2 adsorb on the graphite walls and on the gas/liquid interfaces. As the amount of ethaline inside the pores increases, the graphite surfaces are covered with liquid, and less area is accessible for adsorption of CO_2 at the walls, resulting in a decrease in the average density of CO_2 in the first and second layers, and an increase in the center regions. For rutile pores, CO_2 strongly adsorbs at the walls when little ethaline is present (system TiC1a), but increasing amounts of DES inside the pore leads to preferential adsorption of ethylene glycol at the walls, which displaces CO_2 to the gas/liquid interfaces that form in the second layers and center regions. From our analysis, the number density of carbon dioxide within ethaline is always smaller in the confined DES than in the bulk DES (Fig. 4a); however, when we add the CO_2 adsorbed at the pore walls and at the gas/ethaline interfaces, the number density of carbon dioxide in confined ethaline is significantly enhanced when compared to that in the bulk DES, especially for pores partially filled with ethaline. The results shown in Fig. 4, coupled with the density profiles shown in Fig. 3, indicate that CO_2 adsorption at the pore walls and at the gas/liquid interface are the predominant mechanisms for the increase in CO_2 uptake in nanoconfined ethaline, as compared to what takes place in bulk ethaline. The density profiles of Fig. 3 suggest that the third mechanism described by Ho *et al*.,[27] namely uptake of CO_2 in the low density regions formed by layering of ethaline inside the pore, does not take place in our systems. As expected, the value of the bulk gas pressure also influences the uptake of CO_2 in these systems. For example, for the case of rutile pores partially filled with ethaline, raising the bulk pressure from 4.9 bar (system TiC1) to 16 bar (system TiC1a) leads to an increase in the overall average number density of CO_2, which goes from 5.5 times of the value observed in bulk ethaline (system TiC1, Table S2) to 7.3 times the bulk value (system TiC1a, Table S2 and Fig. 4b). We note that we could have considered systems equivalent to GC1, TiC1 and TiC3 but with bigger pore surface areas and thus larger amounts of adsorbed ethaline, in which we could have accurately quantified the CO_2 dissolved in the nanoconfined ethaline (and thus include these results in Figure 4a). However, we did not perform these calculations because our current results already show that CO_2 adsorption at the pore walls and at the gas/liquid interface are the predominant mechanisms for the increased CO_2 uptake in nanoconfined ethaline, as compared to what is observed for CO_2 in bulk ethaline; the amounts of CO_2 dissolved in confined ethaline are either comparable or smaller to the CO_2 dissolved in bulk ethaline.

We also determined the number density of CO_2 in empty graphite/rutile pores (i.e., without ethaline). In the GC0, GC0a, TiC0 and TiC0a systems, the number density profiles show that CO_2 prefer to stay close to the pore surfaces (Fig. S2g-j), but the snapshots shown in Figs. S1e-h suggest that the interactions between CO_2 and rutile are much stronger than with graphite, in the sense that all CO_2 molecules adsorb at the rutile walls in system TiC0 (Fig. S1g); raising the bulk pressure of CO_2 in system TiC0a (Table 1) leads to increased CO2 adsorption within the rutile pore (Fig. S1h). The calculated average number density of CO_2 in the first layers of these systems range between 4.99-8.87 molecule/nm^3 in the graphite systems GC0 and GC0a, and between 8.23-19.12 molecule/nm^3 in the rutile systems TiC0 and TiC0a (Table S2). If we consider the whole pore volume in these systems, the overall average number density ranges between 1.16-2.66 molecule/nm^3 (carbon pores) and 1.11-5.54 molecule/nm^3 (rutile pores, Table 2). These results and those shown in Fig. 4 suggest that empty graphite or rutile pores can adsorb more CO_2 than nanoconfined ethaline, which in turn can uptake more carbon dioxide than bulk ethaline. These findings are similar to the trends reported by Shi and Sorescu for CO_2 and H_2 in the IL 1-*n*-hexyl-3-methylimidazolium bis(trifluoro-methylsulfonyl)amide, [hmim$^+$][Tf$_2$N$^-$] confined in single-walled carbon nanotubes.[58] Nevertheless, we note that selection of adsorbents for CO_2 capture is not solely based on adsorption properties; other issues have to be considered, such as economics, ease of regeneration, effect of variables such as temperature, pressure, moisture and effects of other contaminants on adsorption properties, etc.[11-13] In addition, in separations of CO_2 using membranes, properties such as solubility, diffusion and permselectivities of the different species need to be considered.[11-13] Furthermore, here we just considered a common DES, ethaline, inside common slit graphite an rutile pores; other DES and other materials with nanopores of different geometries could be screened and explored, looking for optimal systems with enhanced CO_2 separation properties.

The density profiles shown in Figs. 3 and S2 suggest that in all our confined systems, the molar ratio between choline chloride and ethylene glycol exhibits important local variations with respect to the bulk value of 1:2. The ratio choline : chloride : ethylene glycol is reported in the whole pore volume and in the different regions of the pore in Table 2; note that the overall molar ratio in all our systems (pore + reservoirs) is 1:1:2. In analogy to our findings for choline iodide + glycerol inside slit graphitic and rutile pores,[31] confinement and interactions with the pore walls influence the ratio of the ethaline species inside the pore. In all graphite systems, the overall ratio inside the pore is the same as the bulk value, but significant local variations are observed in the different regions as we vary the amount of ethaline inside the pore. For all graphite systems except GC1, ethylene glycol is present in the first layers at more than twice the bulk ratio, while the anions are slightly short of the bulk ratio; however, in the second layers and center regions ethylene glycol and chlorine are slightly in deficit and in excess, respectively, of the bulk ratio. In contrast, in system GC1 ethylene glycol is the dominant species in the center of the pore, as it tends to accumulate near the gas/liquid interface in the center of the pore, and is slightly in deficit in the second layers; meanwhile, the anions are in excess of the bulk ratio in both the second layer and center regions. In all rutile systems, the overall ratio of ethaline species inside the pores is significantly different from the bulk ratio, showing in all cases an excess of ethylene glycol (here we did not account for the ions adsorbed at the edges of the rutile walls, see Figs. 2e-h). The largest difference in overall ratio is observed for system TiC1, which has a large excess of HBD adsorbed on the walls; similar trends are observed for system TiC1a (not shown for brevity). For all rutile systems, ethylene glycol is the absolute dominant component in the first layers, but is short of the bulk ratio in the second layers and center region; in contrast, the anions are slightly in deficit in the first layers and close to the bulk ratio in the second layer and center region. For both graphitic and rutile systems, the presence of carbon dioxide (compare systems GC4-G4, GC8-G8, TiC4-Ti4 and TiC8-Ti8 in Table2) leads to slight changes in the ratio of ethylene glycol (mainly in the first layers but also in the second layers of rutile systems), and does not seem to affect the other ethaline species.

Table 2. Ratio of choline : chloride : ethylene glycol inside the pore area.[a]

Model system	Overall	First layers	Second layers	Center region
GC1	1.0 : 1.0 : 2.0	1.0 : 0.8 : 2.0	1.0 : 1.2 : 1.3	1.0 : 1.7 : 6.5
GC3	1.0 : 1.0 : 2.0	1.0 : 0.7 : 4.3	1.0 : 1.1 : 1.4	1.0 : 1.0 : 1.7
GC4	1.0 : 1.0 : 2.0	1.0 : 0.7 : 4.3	1.0 : 1.1 : 1.4	1.0 : 1.0 : 1.7
GC8	1.0 : 1.0 : 2.0	1.0 : 0.8 : 5.1	1.0 : 1.0 : 1.5	1.0 : 1.0 : 1.8
TiC1	1.0 : 0.8 : 5.8	1.0 : 0.7 : 6.5	1.0 : 1.1 : 0.0	1.0 : 1.0 : 1.7
TiC3	1.0 : 0.9 : 2.4	1.0 : 0.6 : 7.4	1.0 : 1.2 : 0.9	1.0 : 0.9 : 1.3
TiC4	1.0 : 0.9 : 2.3	1.0 : 0.7 : 7.2	1.0 : 1.1 : 1.5	1.0 : 0.9 : 1.3
TiC8	1.0 : 1.0 : 2.2	1.0 : 0.7 : 10.8	1.0 : 1.1 : 2.0	1.0 : 1.0 : 1.6
G4	1.0 : 1.0 : 2.0	1.0 : 0.7 : 4.0	1.0 : 1.1 : 1.4	1.0 : 1.0 : 1.7
G8	1.0 : 1.0 : 2.0	1.0 : 0.8 : 4.9	1.0 : 1.0 : 1.5	1.0 : 1.0 : 1.8
Ti4	1.0 : 0.9 : 2.3	1.0 : 0.7 : 6.2	1.0 : 1.3 : 1.8	1.0 : 0.9 : 1.3
Ti8	1.0 : 1.0 : 2.1	1.0 : 0.6 : 9.2	1.0 : 1.1 : 2.2	1.0 : 1.0 : 1.5
TiC4s	1.0 : 1.0 : 2.3	1.0 : 0.7 : 5.8	1.0 : 1.1 : 2.4	1.0 : 1.0 : 1.7

[a] The bulk DES has a ratio 1.0 : 1.0 : 2.0

3.2. Average interaction energies

The results shown in the previous section can be explained by examining the interaction energies in our systems. Figure 5 (and Table S3, Supporting Information) shows the average interaction energies of CO_2 molecules with cations, anions, ethylene glycol molecules, other CO_2 molecules, and the pore walls in our systems. The values reported in Fig. 5 and Table S3 are the sum of the van der Waals and electrostatic interaction energies, and all CO_2 molecules near ethaline were considered (i.e., adsorbed at the pore walls, dissolved within confined ethaline, and near the gas/liquid interfaces, even if those are outside of the pore volume). For comparison purposes, the average interaction energy of CO_2 with bulk ethaline species are also included in Table S3. In general, for all systems the CO_2-wall interactions diminish with increasing amounts of ethaline in the pores (Fig. 5), with the largest CO_2-wall energies observed in systems without ethaline (GC0, GC0a, TiC0 and TiC0a, see Table S3). In contrast, the interactions of CO_2 with ethaline species increase as the amount of DES in the pores is raised (Fig. 5), and reach maxima in the case of bulk systems (Table S3). The CO_2-wall interactions in rutile systems are much larger than their counterparts in graphite systems that have either no ethaline (TiC0a vs GC0a) or little amount of DES present (TiC1a vs GC1); however, as the amount of ethaline in the pores increases, the CO_2-wall interactions in rutile systems are much smaller than those observed in graphite systems with the same amounts of ethaline (Fig. 5). The interaction energies of ethaline species with the rutile walls (see Table S4, Supporting Information) are much larger than the CO_2-rutile interactions (Table S3), and much larger than the ethaline-wall interactions observed in graphite systems. Therefore, the rutile walls strongly adsorb CO_2 molecules in the absence of DES, but as more ethaline is present in the pore, the rutile walls quickly uptake the DES (mainly ethylene glycol), which displaces CO_2 from the walls. Similar trends are observed in graphite systems, but the process of uptake of ethaline and displacement of CO_2 is more gradual, as the ethaline-wall interactions are weaker and comparable in magnitude with the CO_2-graphite energies (see Tables S3 and S4). The presence of CO_2 seems to have negligible influence on the interactions of ethaline species with the pore walls (compare energies for systems GC4-G4, GC8-G8, TiC4-Ti4 and TiC8-Ti8, Table S4).

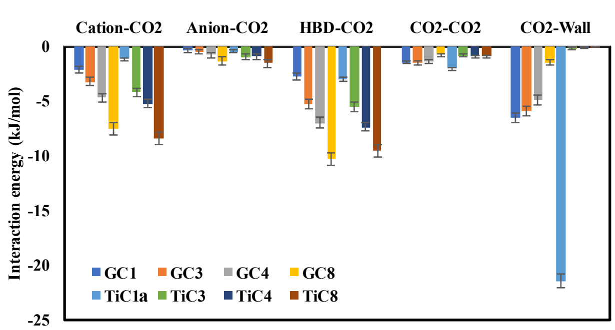

Figure 5. Average interaction energies (electrostatic + van der Waals) cation-CO_2, anion-CO_2, HBD-CO_2 and CO_2-walls in pore systems. See also Tables S3 and S4 (Supporting Information).

3.3. Radial distribution functions

In order to obtain insights into the structure of ethaline and CO_2 confined inside the nanopores, we monitored the radial distribution functions (RDFs) $g(r)$ between the center of mass (COM) of the ions, the HBD ethylene glycol and CO_2, as well as between relevant atoms of the different species. All RDFs in the pore systems are corrected according to the volume, so that they converge to 1 at long distances. In Figures S4-S7 (Supporting Information) we present the COM RDFs and several relevant atom-atom RDFs for all ethaline species and CO_2 confined inside the overall pore volume (i.e., not distinguishing molecules or ions in the different regions of the pore). Results are shown for all pore systems with the exception of TiC1 and TiC3, as these systems are highly inhomogeneous because they contain small amounts of ethaline, and a significant quantity of DES is adsorbed on the edges of the rutile walls. For comparison purposes, COM RDFs and atom-atom RDFs for ethaline species in the bulk system are presented in Figure S8 (Supporting Information). For both graphite and rutile systems, the results displayed in Figs. S4-S7 show that different amounts of ethaline in the pores, as well as presence/absence of CO_2, only lead to subtle differences in these RDFs; almost all of the peaks, shoulders and other features in the $g(r)$ functions are observed around similar values of r, with only observable differences in signal intensity (i.e., the height of the peaks and shoulders). One very subtle difference is observed in the COM RDF between anion and ethylene glycol for TiC4 and Ti4 systems (Fig. S5e), where a small peak appears at $r \sim 0.24$ nm that is not present for bulk ethaline (Fig. S8b). In rutile systems, the RDF between atoms OD and HOD from ethylene glycol (see Fig. 1 for atom notation) shows a small peak at $r \sim 0.44$ nm for systems TiC4 and Ti4 (Fig. S7d), which is not present either in systems TiC8/Ti8 or in bulk ethaline (Fig. S8d).

Additional differences are observed if we examine the COM and atom-atom RDFs, now distinguishing between species in the different layers/regions of the pore. For simplicity and brevity, only the results for systems GC8 and TiC8 are presented in Figures S9-S12 (Supporting Information; noisy signals involving CO_2 molecules are mainly caused by the small amount of carbon dioxide present in these systems); selected RDFs that exhibit differences are summarized in Figures 6 and 7. In analogy to the results shown in Figs. S4-S7 for ions/molecules in the overall pore volume, the local RDFs (Figs. S9-S12) do not show big differences between systems with different ethaline loadings, or with/without CO_2. For graphite system GC8 (Fig. 6), the main differences are observed for ethaline components in the second layers: (1) for RDF between cations (Fig. 6a), the valley between the first and second peak almost vanishes; (2) for RDF between cations and anions (Fig. 6b), the second peak disappears; (3) for RDF between cations and ethylene glycol (Fig. 6c) the first peak moves from 0.52 nm to 0.5 nm; (4) for RDF between anions and ethylene glycol (Fig. 6d), a peak at 0.24 nm appears (this peak also appears in the first layers and center region). No significant differences are observed in the atom-atom RDFs between the different layers in the graphite system (Fig. S11), except for distinct signal intensities. For the rutile system TiC8, the main differences in RDFs (Fig. 7) are observed in the first layers, mainly due to the strong interactions exerted by the rutile walls. The structure of the first layer near the walls is very different from that of the second layers and center regions, and in turn very different from that of bulk ethaline, as indicated by the COM RDFs between the cations (Fig. 7a), cation and ethylene glycol (Fig. 7c), and ethylene glycol molecules (Fig. 7e). Furthermore, atom-atom RDFs between OD in ethylene glycol and HO1 in choline (Fig. 1), and with HOD and HOE in ethylene glycol (Fig. 1), show significant differences in the first layers as compared to the RDFs in the second layers and center regions (Figs. 7f-h). These results indicate that strong interactions exerted by the rutile walls cause significant changes in the structure of ethaline in the first layers.

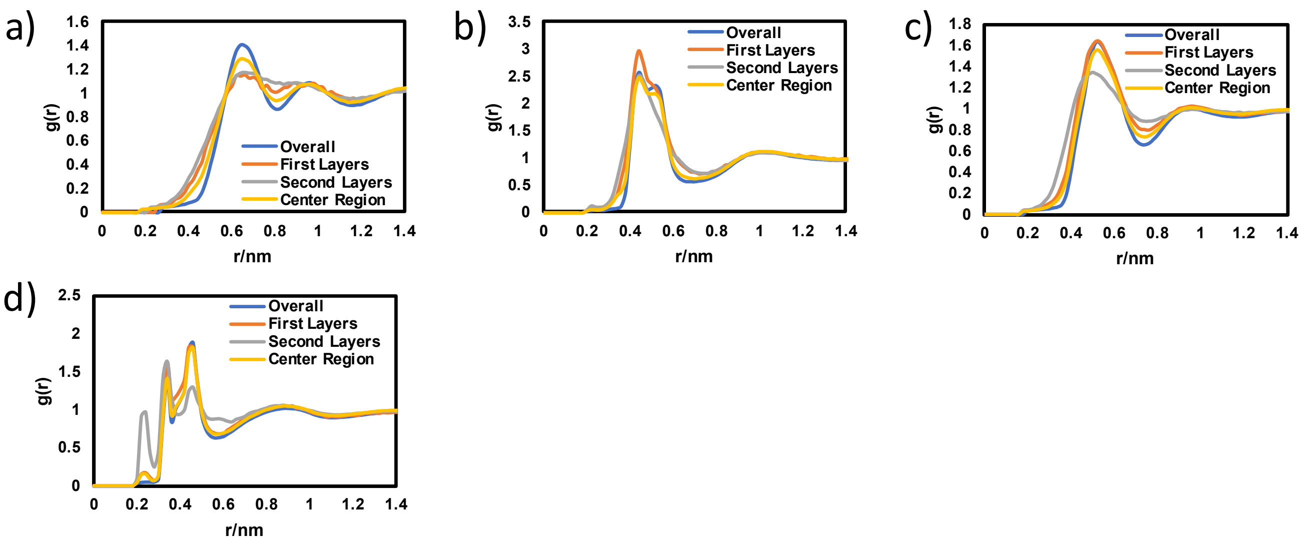

Figure 6. Selected COM RDFs for ethaline species in different regions in graphite system GC8: (a) Cation-Cation, (b) Cation-Anion, (c) Cation-HBD, (d) Anion-HBD.

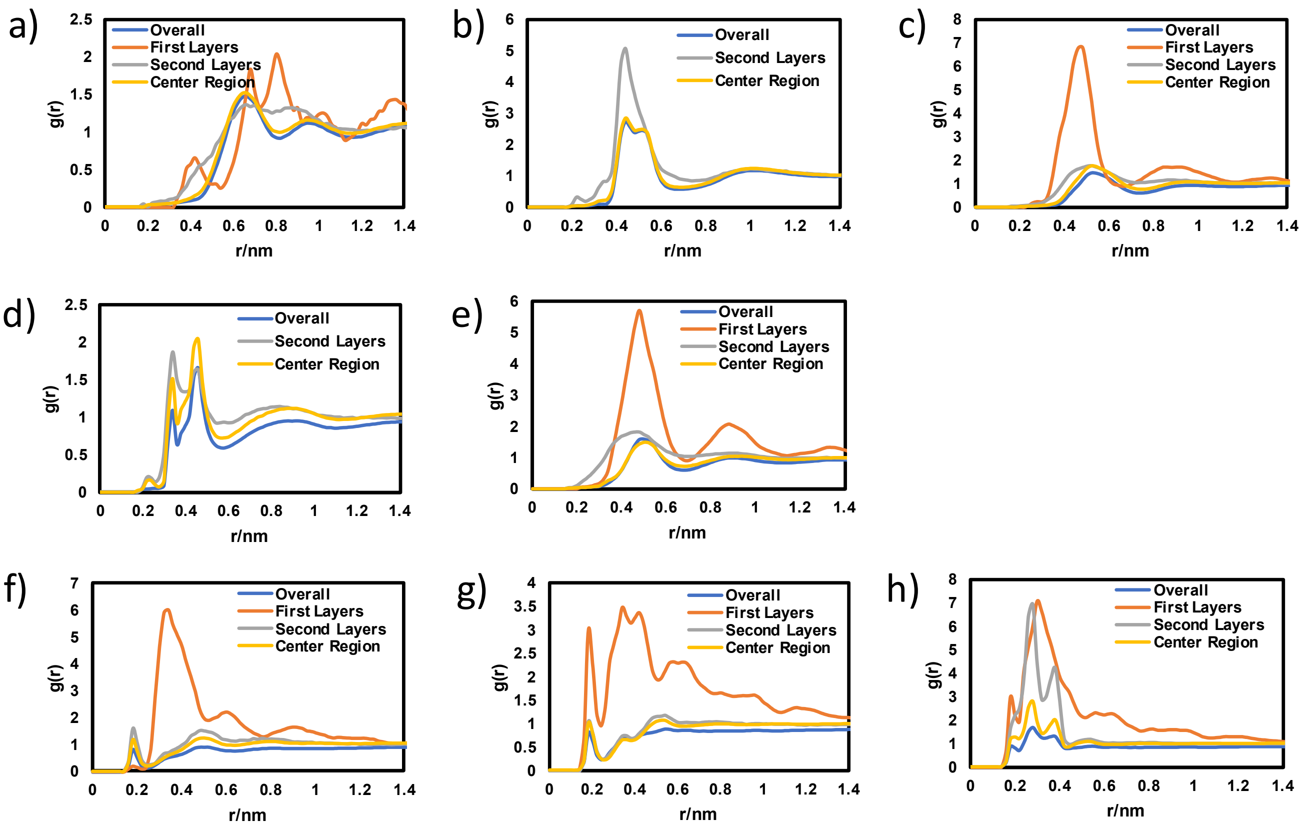

Figure 7. Selected COM RDFs and atom-atom RDFs for ethaline species in different regions in rutile system TiC8: (a) Cation-Cation, (b) Cation-Anion, (c) Cation-HBD, (d) Anion-HBD, (e) HBD-HBD, (f) OD(HBD)-HO1(cation), (g) OD(HBD)-HOD(HBD), (h) OD(HBD)-HOE(HBD).

3.4. Diffusion coefficients of ethaline and CO_2

Overall diffusion coefficients of ethaline species in our model systems are presented in Table 3; results obtained in the bulk(C) system are also included. In the graphite systems, the overall diffusion coefficients of ethaline species tend to diminish with increasing loading of ethaline inside the pores (compare systems GC1-GC3-GC4-GC8, and G4-G8 in Table 3). At small ethaline loadings, a larger portion of DES is at the liquid/gas interfaces and thus the species have more freedom to move around. This effect is especially noticeable for ethylene glycol as it is the predominant species at the liquid/gas interfaces, and experiences the largest reduction in diffusion coefficient with increasing loadings of DES inside the pore. If only ethaline species inside the pore volume are considered (i.e., we do not account for ethaline near the menisci in systems GC8 and G8, see Figs. 2d and S1b), the diffusion coefficients of ethaline species are slightly smaller than those observed when the menisci are included (Table 3). When carbon dioxide is present in graphite systems, the diffusion coefficients for all ethaline species are larger compared to those observed in similar systems that do not contain CO_2 (compare systems GC4/G4 and GC8/G8 in Table 3). In contrast, in rutile systems the presence of carbon dioxide molecules does not seem to have a strong influence on the diffusion coefficients of ethaline species (compare systems TiC4/Ti4 and TiC8/Ti8 in Table 3). Furthermore, in system TiC1 the ethaline components are strongly adsorbed onto the rutile walls, as reflected by the very small diffusivities observed. Increasing the amount of ethaline inside the rutile pores leads to increases in the diffusion coefficients of the DES species, with the exception of ethylene glycol from system TiC4/Ti4 to system TiC8/Ti8 (Table 3). This observation might be related to the strong spatial inhomogeneities observed for systems TiC4/Ti4 (Figs. 2g and S1c), where the HBD is the main component present at the gas/liquid interfaces, and thus is more mobile compared to the situation where ethaline completely fills the rutile pores (systems TiC8/Ti8, Figs. 2h and S1d)

Table 3. Overall diffusion coefficients (10^{-11} m^2/s) of ethaline species in our model systems.

Model system	Cation	Anion	HBD
Bulk(C)	3.81 ± 0.07	4.75 ± 0.02	7.40 ± 0.04
GC1	16.50 ± 0.08	15.92 ± 0.08	45.36 ± 0.08
GC3	10.43 ± 0.02	11.75 ± 0.02	20.41 ± 0.03
GC4	7.0 ± 0.5	8.2 ± 0.5	17.9 ± 0.5
GC8	3.9 ± 0.4	4.7 ± 0.5	10.4 ± 0.6
GC8(*)	3.5 ± 0.1	4.1 ± 0.1	6.8 ± 0.1
G4	3.4 ± 0.1	4.9 ± 0.1	11.0 ± 0.1

G8	3.6 ± 0.3	4.7 ± 0.3	15 ± 1
G8(*)	2.9 ± 0.2	3.8 ± 0.2	6.1 ± 0.3
TiC1	0.003 ± 0.002	0.004 ± 0.001	0.199 ± 0.002
TiC3	0.22 ± 0.02	0.44 ± 0.01	2.30 ± 0.02
TiC4	1.42 ± 0.03	1.97 ± 0.02	5.6 ± 0.1
TiC8	1.86 ± 0.03	2.73 ± 0.02	5.08 ± 0.3
TiC8(*)	1.83 ± 0.04	2.49 ± 0.02	3.25 ± 0.01
Ti4	1.42 ± 0.08	2.2 ± 0.1	5.9 ± 0.2
Ti8	1.84 ± 0.01	2.59 ± 0.02	6.12 ± 0.07
Ti8(*)	1.76 ± 0.01	2.48 ± 0.01	3.33 ± 0.01

(*) Only ethaline components within the pore considered for the measurement (i.e., components in the menisci outside of the pore volume were not considered). In the case of rutile systems, only ions/molecules in the center 8-nm range in the *x*-direction of the pore were considered for the analysis to remove end-wall effects.

Overall diffusion coefficients of CO_2 inside pore systems are presented in Figure 8; data is reported in Table S5 (Supporting Information). In our pore systems, carbon dioxide molecules have much faster dynamics comparing to ethaline, and therefore move in and out of the pore volume frequently. To obtain the results reported in Figure 8, we only consider the CO_2 molecules that stay inside the pore volume during the whole (or most of the) simulation time, and therefore these results have larger uncertainties when compared to the results obtained for ethaline in Table 3. For systems GC8 and all the rutile systems, we also accounted for CO_2 molecules near the gas/liquid interfaces even if those are outside of the pore volume. The diffusion coefficient of CO_2 inside the bulk(C) system is only slightly larger to the diffusivities observed in systems GC8 and TiC8 that are completely filled with ethaline, suggesting that confinement in our 5.2 nm pores does not seem to have a strong effect on the diffusion of CO_2 (Fig. 8, Table S5). In pores that do not contain ethaline (systems GC0 and TiC0), CO_2 adsorbs at the pore walls and have stronger interactions with rutile than with graphite, which makes the diffusivity of CO_2 to be smaller in system TiC0 than in system GC0. Raising the bulk gas pressure (systems GC0a and TiC0a) lead to an increase in the diffusivity of CO_2 inside the pores, as the additional molecules of carbon dioxide are not as strongly bound to the pore walls as it was the case when the pressure was lower (compare systems GC0, GC0a, TiC0 and TiC0a in Table S5). CO_2 diffuses ~30 times faster in graphite pores without ethaline as compared to in pores fully filled with DES, or in bulk ethaline (Fig. 8, Table S5); similar trends are observed for rutile pores, but the diffusivities of CO_2 in empty rutile pores are just about 3.6 faster compared to pores filled with ethaline. These trends are similar to the results reported by Budhathoki *et al.*,[24] where the mobility of carbon dioxide was more than 100 times slower inside graphite pores completely filled with the IL $[C_4mim^+][Tf_2N^-]$ than in empty graphite pores. Shi and Sorescu[58] also reported that carbon dioxide molecules diffuse significantly faster in empty carbon nanotubes than in nanotubes partially filled with the IL $[hmim^+][Tf_2N^-]$; CO_2 had the slowest diffusivity when dissolved in the bulk IL.[58] As the amount of ethaline in the pores increases, CO_2 is displaced from the walls to the gas/liquid interfaces and has increased mobility, resulting in faster CO_2 diffusion compared to the cases where no ethaline is present, and to when the pores are completely filled with ethaline (compare systems GC1 with GC0 and GC8, and TiC1a with TiC0a and TiC8 in Fig. 8 and Table S5). Further increases in the amount of ethaline inside the graphitic pores lead to a reduction in CO_2 diffusion (compare systems GC1-GC3-GC4-GC8 in Figs. 8 and Table S5). In contrast, diffusivity of CO_2 in system TiC3 is the largest, but further increases in the amount of ethaline lead to slower CO_2 diffusion. This observation might be caused by the fact that system TiC3 is very inhomogeneous, with two ‘bridges’ of liquid ethaline enclosing a gas-like phase within the pore.

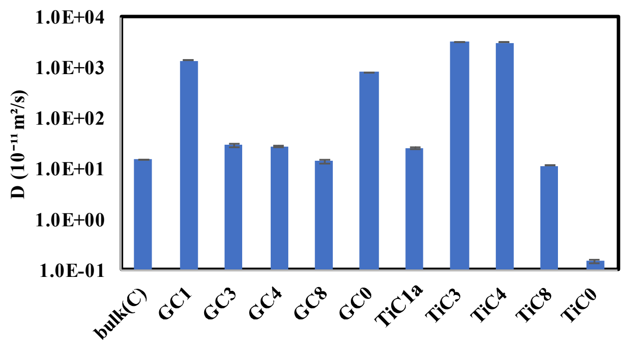

Figure 8. Overall diffusion coefficients of CO_2 in our model systems. Data reported in Table S5 (Supporting Information)

4. Conclusions

Molecular dynamics simulations were performed to study the behavior of carbon dioxide in varying amounts of ethaline (a typical deep eutectic solvent formed by mixing choline chloride and ethylene glycol with a molar ratio of 1:2), when confined in slit-like graphite or rutile nanopores of width H = 5.2 nm at T = 318 K. In the absence of ethaline, CO_2 adsorbs to the graphite and rutile walls, but increasing amounts of ethaline inside the pores quickly displace CO_2 from the walls and into the gas/liquid interfaces, and into dissolution within the confined ethaline. This process is very pronounced for rutile pores, which have very strong interactions with the ethaline components. The strength of the CO_2-wall interactions decrease as more ethaline is adsorbed by the slit pores, while the interactions between the CO_2 molecules and ethaline components become stronger as more DES is present inside the pores. Strong interactions with the pore walls also cause important local variations in the ratio of choline, chloride and ethylene glycol, especially in rutile systems. Interactions with the rutile walls also cause important changes in the liquid structure of the first layers of ethaline that are closest to the walls; however, the overall liquid structure of the confined DES is mostly similar to that of bulk ethaline regardless of the pore walls, varying amounts of ethaline or presence of CO_2 in these systems. Our results indicate that the overall diffusion coefficients of the ethaline components mostly diminish with increasing loadings of DES inside the pores. Presence of CO_2 increases the mobilities of ethaline components in graphitic pores, when compared to systems without CO_2; in contrast, the diffusivities of ethaline components in rutile pores are not significantly affected by the presence of carbon dioxide. CO_2 diffuses faster in pores without ethaline as compared to in pores fully filled with DES, or in bulk ethaline. As the amount of ethaline in the pores increases, CO_2 is displaced from the walls to the gas/liquid interfaces and has increased mobility, resulting in faster CO_2 diffusion compared to the cases where no ethaline is present, and to the situation where the pores are completely filled with ethaline.

Our results also indicate that the average number density of CO_2 dissolved within confined ethaline is smaller than what is dissolved in bulk ethaline. However, when we account for all CO_2 present in our confined systems (i.e., dissolved in confined ethaline, and adsorbed at the pore walls and at the gas/liquid interface), the average number density of CO_2 is significantly larger than what is observed in the bulk DES. For pores partially filled with ethaline, the overall average number density of CO_2 in graphite systems can be ~3-4 times the value observed in bulk ethaline, and up to 7.3 times the bulk value for the case of rutile systems. We note, however, that pores with no ethaline exhibit even larger average number densities of carbon dioxide, suggesting that empty graphite or rutile pores can adsorb more CO_2 than nanoconfined ethaline, which in turn can uptake more carbon dioxide than the bulk DES. However, we also note that selection of adsorbents or membranes for CO_2 separations are not solely based on adsorption properties, but a complex combination of factors such as economics, ease of regeneration, presence of moisture or other contaminants, and properties such as solubility, diffusion and permselectivities of the different species present. Furthermore, here we only considered a common DES, ethaline, inside common slit graphitic and rutile pores. Other DESs, ILs and solvents, as well as additional porous materials with different pore geometries and pore chemistries could be analyzed and screened, for example using computational tools, aiming at finding a few optimal candidate systems with enhanced CO_2 separation and adsorption properties. These candidate systems could then be studied in further detail, using a combination of MD simulations and Monte Carlo techniques (e.g., Gibbs ensemble MC, grand canonical MC, continuous fractional component MC, all coupled with configurational bias methods, as described in refs. 24, 47-52). We intend to explore these ideas in our future studies. Overall, our results suggest that pores partially filled with DESs can be used for separation of carbon dioxide, in analogy to systems of supported ionic liquid membranes and supported ionic liquid phase materials.

Supporting Information Available

Data for number density of carbon dioxide, average interaction energies and diffusion coefficients; additional simulation snapshots, local number density profiles, and radial distribution functions. This information is available free of charge via the Internet at http://pubs.acs.org/.

Acknowledgements

This work was partially supported by the National Science Foundation (CAREER Award CBET-1253075), and by Northeastern University. High-performance computational resources for this research were provided by Research Computing at Northeastern University (https://www.northeastern.edu/rc/), High Performance Computing at Louisiana State University (http://www.hpc.lsu.edu), and the Louisiana Optical Network Initiative (http://www.loni.org).

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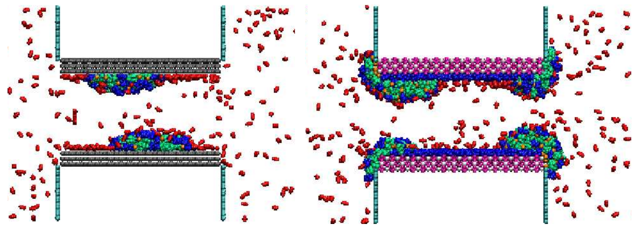

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